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Next item →

1. If f is a bijective function, then $f^{-1} \circ f$ is equal to the identity function.

1 / 1 point

- ☒ True
- ☐ False

👍 Nice work

$$(f^{-1} \circ f)(x) = (f \circ f^{-1})(x) = x$$

2. Given a function $f : \mathbb{R} \rightarrow [0, \infty)$ with $f(x) = x^2$, then:

1 / 1 point

- ☐ $f^{-1}(x) = \sqrt{x}$
- ☐ $f^{-1}(x) = -\sqrt{x}$
- ☒ f^{-1} doesn't exist as f is not a bijection
- ☐ $f^{-1}(x) = \frac{x}{2}$

👍 Nice work

f is not injective, hence, it is not a bijection

3. Given a function $f : [0, \infty) \rightarrow [4, \infty)$ with $f(x) = 3x + 4$, then:

1 / 1 point

- ☒ $f^{-1}(x) = \frac{x-4}{3}$
- ☐ $f^{-1}(x) = \frac{x+4}{3}$
- ☐ $f^{-1}(x) = \frac{x-3}{4}$
- ☐ f^{-1} doesn't exist as f is not a bijection

👍 Nice work

f is a bijection. $f(x) = y \implies 3x + 4 = y \implies x = \frac{y-4}{3}$ hence, $f^{-1}(x) = \frac{x-4}{3}$

4. Which one of the following is the inverse function $f(x) = (x + 4)^2, x \geq 0$?

1 / 1 point

- ☒ $f^{-1}(x) = \sqrt{x} - 4$
- ☐ $f^{-1}(x) = \sqrt{x} + 4$
- ☐ $f^{-1}(x) = \sqrt{x - 4}$
- ☐ $f^{-1}(x) = \sqrt{x + 4}$

👍 Nice work

$f(x) = y \implies (x + 4)^2 = y \implies x + 4 = \sqrt{y} \implies x = \sqrt{y} - 4$ hence,
 $f^{-1}(x) = \sqrt{x} - 4$

5. The following graph contain the curves of three functions, f , g and h . Which one of these three functions is invertible?

1 / 1 point

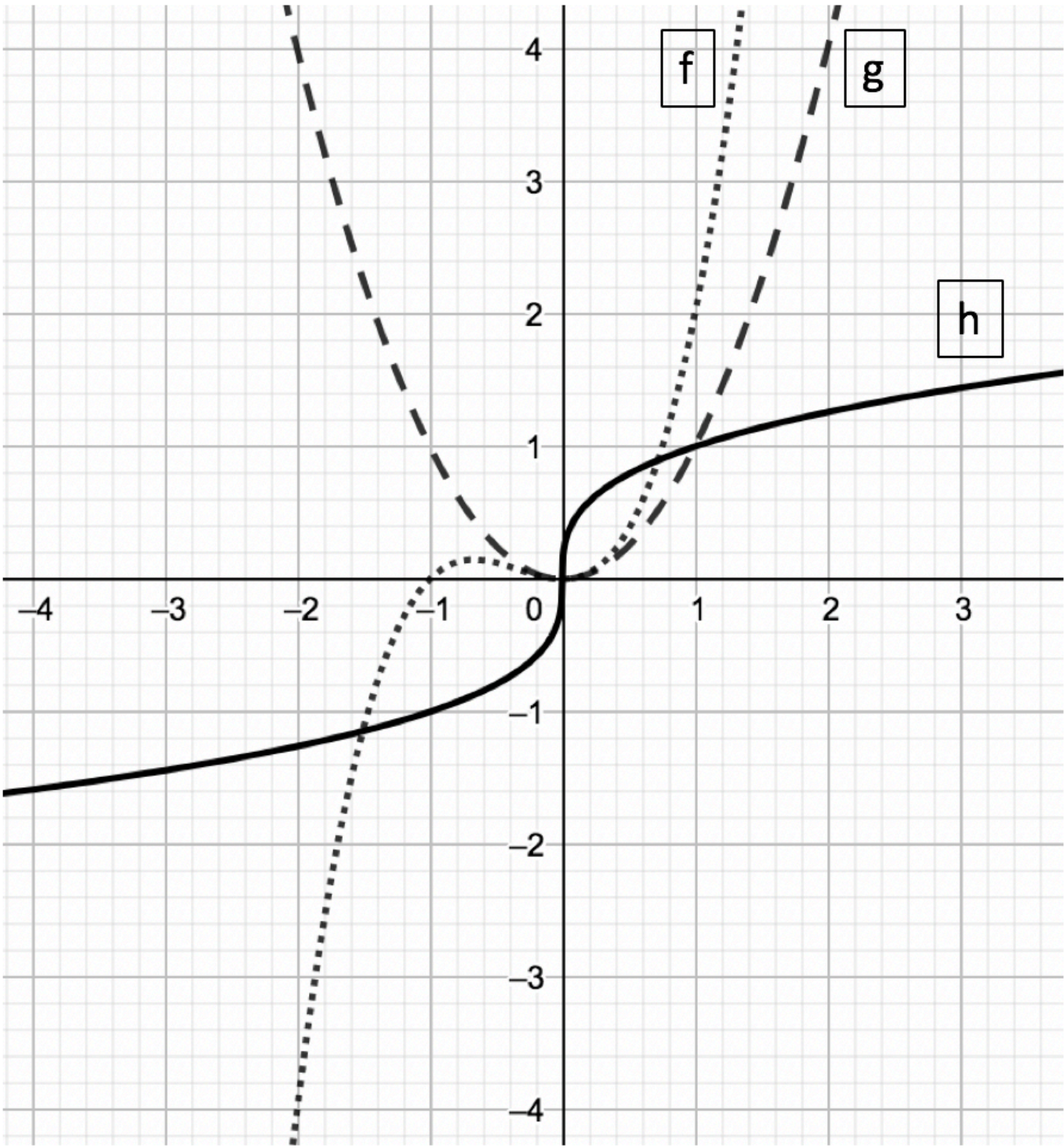


Image description:

The domain of all three functions is $\mathbb{R}$.

The curve for h is strictly increasing from $-\infty$ to $+\infty$.

The curve for f crosses the x -axis on $x = -1$ and on $x = 0$.

The curve for the function g is symmetric with respect to the straight line $x = 0$.

- ☒ h
- ☐ g
- ☐ f
- ☐ None of these answers

👍 Nice work

The curve of the function h is a bijection whereas the other curves are not.